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7 August 1987
SUPERSEDING
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25 August 1981

MILITARY SPECIFICATION

FILTER-SEPARATOR, LIQUID FUEL:

FRAME MOUNTED, 600 GPM CAPACITY (2271 LITER PER MINUTE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 600-gallon per minute (gpm) (2271 liter per minute) capacity filter-separator for use in removing undissolved water and solid contaminants from petroleum fuels (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-F-8901	- Filter-Separators, Liquid Fuel; and Filter-Coalescer Elements, Fluid Pressure: Inspection Requirements and Test Procedures for.
MIL-W-22248	- Weldments, Aluminum and Aluminum Alloy.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-F-52308	- Filter Element, Fluid Pressure.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13217E9340	- Filter-Separator, Liquid Fuel: Frame Mounted, 600 GPM (2271 liter) Capacity.

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted (adopted documents are marked with an asterisk (*)) shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ALUMINUM ASSOCIATION (AA)

Aluminum Construction Manual, Section 1, Specification for Aluminum Structures, 7.2 Welding Fabrication.

(Application for copies should be addressed to the Aluminum Association, 420 Lexington Avenue, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

* B 633 - Electrodeposited Coatings of Zinc on Iron and Steel.
* D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33125.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The filter-separator shall be in accordance with top assembly TA13217E9340 and as specified herein. The filter-separator shall consist essentially of a vertical aluminum vessel with removable cover, a manually operated air vent, inlet and outlet connections, filter elements, canisters, a differential pressure gage, a pressure-relief valve, a water-level sight gage, a ground rod, a ground cable, attaching hardware, and an automatic water-drain valve combined with two gate valves for manual operation. The filter-separator shall be mounted within a tubular rectangular frame. The general requirements, operating and environmental conditions, shall be in accordance with MIL-F-8901. The filter-separator shall have a rated flow of 600 gpm (2271 liters) and a maximum working pressure of 150 pounds per square inch (psi) (10.34 kPa).

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the filter-separator. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.2). Any changes or deviations of the filter-separator from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish filter-separators conforming to this specification.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings.

3.3.1 Material deterioration prevention and control. The filter-separators shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the filter-separators may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

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3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the filter-separaters may be newly fabricated from recovered materials to the maximum extent practicable, provided the filter-separators produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the filter-separators.

3.5 Environmental conditions.

3.5.1 Operating temperature. The filter-separator shall permit rated flow in any ambient temperature in the range of +125 °F (+52 °C) to -25 °F (-32 °C).

3.5.2 Storage temperatures. The filter-separator shall not be damaged by storage at ambient temperatures from +160 °F (+71 °C) to -50 °F (-46 °C).

3.6 Human factors. The filter-separator shall conform to the applicable human factors engineering criteria specified in MIL-STD-1472.

3.7 Maintainability.

3.7.1 Maintenance ratio. The filter-separator shall have a maintenance ratio of no greater than 0.06.

3.8 Reliability. The specified mean-time-between-failure (MTBF) shall be 140 hours when the filter-separator is operated at rated capacity and tested as specified in 4.5.2.5.

3.9 Transportability. The filter-separator shall withstand the dynamic vibrational stresses encountered in transportation and service. The filter-separator shall operate as specified herein when tested as specified in 4.5.2.6.

3.10 Performance.

3.10.1 Vessel (tank). The filter-separator vessel shall be fabricated of aluminum alloy as specified on the drawings. The vessel shall withstand a hydrostatic pressure of 225 psi (.15 kPa) for not less than 30 minutes without leakage, permanent deformation or other defects that could harmfully affect the performance and serviceability of the filter-separator.

3.10.1.1 Deck plate. The vessel deck plate shall be fabricated of aluminum alloy and of the thickness as specified on the drawings. The deck plate shall withstand a minimum of 75 psi (5.17 kPa) without leakage, permanent deformation or other defects that could harmfully affect the performance and serviceability of the filter-separator.

3.10.2 Differential pressure. The maximum differential pressure shall not exceed that specified by MIL-F-8901, table III (see 4.5.2.1.c).

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3.10.3 Permanent separator stage. The permanent separator stage shall be as specified by MIL-F-8901, table III. The direction of flow shall be inside-to-outside (see 4.5.2.1.b).

3.10.4 Water removal. The filter-separator shall remove water as specified by MIL-F-8901, table III from influent fuel containing up to 5 percent (by volume) water (see 4.5.2.1.d).

3.11 Government-furnished property. The media shall consist of 30 filter elements conforming to MIL-F-52308, and 30 canisters as shown on the applicable drawings. The Government will furnish 60 filter elements (NSN 4330-00-983-0998) for each filter-separator (see 6.5); 30 to be installed in the filter-separator and 30 as replacements.

3.12 Differential pressure dial gage. The differential pressure gage shall conform to the dimensions shown on the requirements specified on the applicable drawing. The gage shall provide direct reading of the differential pressure across the filter-separator without manipulation of valves or substraction of readings and shall perform without structural failure or leakage. The accuracy of the gage shall be such that any error shall not exceed ±2 psi (0.001 kPa).

3.13 Hardware. Bolts, nuts and washers shall be fabricated from materials as specified on the applicable drawings. When ferrous materials are specified, the bolts, nuts and washers shall be cadmium plated in accordance with QQ-P-416, type III, class 2 or zinc coated in accordance with ASTM B 633, finish type IV, service condition SC-4. Threads shall conform to FED-STD-H28.

3.14 Valves.

3.14.1 Pressure relief. The pressure-relief valve shall be set to open at a pressure of 170 psi (.12 kPa), ±5 psi (.003 kPa). The valve shall be mounted on the filter-separator with the exhaust opening directed toward the ground. The valve shall not leak when in the closed position.

3.14.2 Automatic water drain. The automatic water-drain valve shall be actuated to drain when the water rises to not more than 2-1/16 inches (5.24 cm) above the bottom of the canister skirt. When the water level has receded a full 1-1/2 inches (3.81 cm), the valve shall automatically close.

3.15 Cover gasket replacements. Four extra O-rings, as shown on the applicable drawings, shall be furnished with each filter-separator as replacement items.

3.16 Float-control gasket replacement. One extra gasket, as shown on the applicable drawings, shall be furnished with each filter-separator as a replacement item.

3.17 Sight-gage gasket replacement. One extra gasket, as specified on the applicable drawings, shall be furnished with each filter-separator as a replacement item.

3.18 Identification marking. The filter-separators shall be identified in accordance with MIL-STD-130. The identification and instruction plates and markings shall be in accordance with the applicable drawings.

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3.19 Treatment and painting. The exterior of the filter-separator normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), the top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.19.1 Chemical coating. The interior of the filter-separator vessel and all components fabricated from aluminum alloy, which are in direct contact with the petroleum fuels, shall be anodized in accordance with MIL-A-8625, type II, class 1, or coated in accordance with MIL-C-5541, class 1A.

3.20 Cleaning and drying after tests. At the conclusion of all tests, and after the removal of the canisters and elements, the filter-separators shall be wiped clean and air-dried at room temperature to exhaust all traces of fuel and water.

3.21 Workmanship. The filter-separators shall be free from defects such as misalined components, incomplete welds, cracks, burns, leaks, and other defects that could impair the operation and serviceability of the filter-separator. All parts shall be clean and free from dirt, sand, grease, oil, and metal chips. Nonfunctional sharp edges and projecting points, which might present a hazard to personnel, shall be avoided.

3.21.1 Welders and welding.

3.21.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the following codes:

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
AWS D1.1, Structural Welding Code, Section 5, Qualifications.
AA, Aluminum Construction Manual, Section 1, Specifications for Aluminum Structures, 7.2 Welding Fabrication.

Certification shall be made available for review by the contracting officer or designated representative.

3.21.1.2 Welding (aluminum). Welding performed in the fabrication of filter-separators shall be in accordance with MIL-W-22248. Classes of welds shall be as specified on the drawings. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with all referenced standards, specifications, and drawings as applicable.

4.1.3 Disassembly inspection. Failure of any examination or test by the first article filter-separator shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-furnished property shall be excluded from the requirements of this paragraph.

4.2 Classification of inspection. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article filter-separator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The filter-separator shall be tested as specified in 4.5.2.1, a, b, c, d, and 4.5.2.6. When specified (see 6.2), it shall also be tested in accordance with any or all of 4.5.2.2, 4.5.2.4, 4.5.2.5, 4.5.2.7, 4.5.2.8, and 4.5.2.9. Failure of any test shall be cause for rejection or shall be cause for performing the inspection specified in 4.1.3.

4.4 Quality conformance inspection.

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4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each filter-separator shall be tested as specified in 4.5.2.1 a and b. Failure of either test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Filter-separators shall be examined as specified herein for the following defects:

Major

101. Any dimensions and tolerances not as specified.
102. Materials not as specified.
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes, or treatment.
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the filter-separator.
107. Hardware such as bolts, nuts, washers, and screw threads not as specified.
108. Aluminum parts and components not coated as specified.
109. Welding and castings not as specified.
110. Filter-separators not cleaned and air-dried after testing.
111. Workmanship not as specified.
112. Treatment and painting not as specified.
113. Color not as specified.
114. Four O-rings missing or not as specified.
115. One extra float-control gasket missing or not as specified.
116. One extra sight-gage gasket missing or not as specified.
117. Welders not as specified.
118. Welders certification not available.

Minor

201. Identification marking incorrect, illegible or missing.
202. Instruction plates not as specified.

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4.5.2 Tests.

4.5.2.1 Performance. The filter-separators shall be tested in accordance with the following performance requirements specified in MIL-F-8901, table III:

a. Hydrostatic pressure tests.
b. Permanent separator stage (inside-to-outside design).
c. Differential pressure.
d. Water removal, 10 percent.

Nonconforance to 3.10.1, 3.10.1.1, 3.11, or to the applicable requirement of MIL-F-8901 shall constitute failure of this test.

4.5.2.2 Environmental conditions.

4.5.2.2.1 High temperature. The filter-separator shall be tested as specified in MIL-STD-810, method 501.2, procedure I. The maximum test temperature for storage shall be 160 °F, (+71 °C) and the maximum operating temperature shall be 125 °F (+52 °C). The operating period of procedure II shall be either 1 hour or the operating time required to check for any component failure due to high temperature.

4.5.2.2.2 Low temperature. The filter-separator shall be tested as specified in MIL-STD-810, method 502.2, procedure I. The minimum test temperature for storage shall be -50 °F, (-46 °C) and the minimum operating temperature shall be -25 °F (-32 °C). The operating period of procedure II shall be either 1 hour or the operating time required to check for any component failure due to low temperature.

4.5.2.2.3 Failure criteria. Nonconformance to 3.5 or to the applicable requirements of MIL-STD-810 shall constitute failure of either temperature tests.

4.5.2.3 Human factors. The filter-separator shall be evaluated to determine conformance to 3.6. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.4 Maintainability.

4.5.2.4.1 Maintenance ratio. The maintenance ratio shall be computed during first article testing to determine the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Nonconformance to 3.7.1 shall constitute failure of this test.

4.5.2.5 Reliability. The first article filter-separator(s) shall be tested as specified in 4.5.2.1 and 4.5.2.2. At the completion of the above testing, continue testing in accordance with 4.5.2.1d until the separator(s) have accumulated 550 hours of test time. Six or more failures shall be cause for rejection. A failure is defined as any malfunction which causes or may cause:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the filter-separator by continued operation.
c. Safety hazards to personnel.

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Nonconformance to 3.8 shall constitute failure of this test. Dependent failure or malfunctions occurring as the result of improper maintenance procedures or operator error are excluded from consideration as chargeable failures. Replacement of filter elements shall not be considered a failure.

4.5.2.6 Transportability. The filter-separator shall be transported by 2-1/2 ton cargo truck over cross-country at an average speed of 15 mph (24 kph) for a total of 100 miles (161 kilometers). Damage to the filter-separator or components or nonconformance to 3.9 shall constitute failure of this test.

4.5.2.7 Differential pressure dial gage. Subject the gage to a hydrostatic pressure equal to the maximum scale pressure, and hold at this pressure for a minimum of 30 minutes. Within 10 minutes after the pressure is released and without recalibration or adjustment, readings shall be taken at three intervals on the scale; the anticipated initial operating pressure, the pressure at which the elements are changed, and the maximum scale pressure. The gage shall be tapped sharply with the finger at approximately the center of the scale area before each reading. A dead weight tester or mercury manometer with air pressure or calibrated check gage shall be used in calibrating the gage. Testing of the gage shall be in a temperature of 20 $\pm$5.6 °C (68 $\pm$10 °F). Nonconformance to 3.12 shall constitute failure of this test.

4.5.2.8 Pressure relief valve. Subject the valve to a rising pressure of up to 170 psi (.12 kPa) $\pm$5 psi (.003 kPa). When the valve opens, make note of the pressure reached. Repressurize the valve to 160 psi (.11 kPa) and hold for 5 minutes. Check for leaks, then release the pressure. Nonconformance to 3.14.1 shall constitute failure of this test.

4.5.2.9 Automatic and manual water-drain valves. During the water removal test specified in 4.5.2.1d, observe the water-level sight gage and ascertain that the automatic drain valve opens when the water reaches the top mark on the sight gage and closes when the water recedes to the lower mark on the sight gage. Change from automatic to manual drainage by operating the two gate valves. Nonconformance to 3.14.2 or any leakage during operating shall constitute failure of this test.

4.6 Inspection comparison. The Government may select filter-separators at any time during the contract production period and subject these filter-separators to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on filter-separators selected at random from those which have been accepted by the Government and will not include the previously inspected first article filter-separators. In addition to any test specified as part of the inspection comparision, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.6.1 Inspection failure. Failure of an inspection comparison filter-separator to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison filter-separator(s) and shall be cause for refusal by the Government to continue acceptance of production filter-separators until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on filter-separators previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all filter-separators accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, level II.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following major defects. AQL shall be 1.0 percent defective.

119.	Canisters and filter elements not installed.
120.	Inlet and outlet connections not sealed with a cap as specified.
121.	Drain valve not open.
122.	Float control, gasket and tubing not removed.
123.	Opening not protected as specified.
124.	Extra elements not preserved by one of the methods specified.
125.	Extra O-ring, sight-gage gasket and float-control gaskets not preserved as specified.
126.	Consolidation of components not as specified.
127.	Components not secured within the box in a manner to prevent free movement and damage as specified for level A.
128.	Shipping container not as specified for level A.
129.	Commercial packing not in accordance with the referenced document.
130.	Marking missing, illegible, incorrect, or incomplete for level A or commercial.

5. PACKAGING

5.1 Preservation. The filter-separator, with canisters and filter elements installed, shall not require preservation (see 6.6). The manual water-drain valve shall be opened. A plastic cap shall be installed over the inlet and outlet connections. The float control, with ball and gasket, shall be removed and the opening covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and secured in place with tape conforming to PPP-T-60, type IV.

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5.1.1 Extra elements. Each extra filter element shall be preserved by one of the following methods:

a. Each filter element shall be preserved in accordance with MIL-P-116, method IA-13.
b. Each filter element shall be inserted in a polyethylene bag, fabricated from material conforming to L-P-378, type I, class 1, thickness .0040, and closed by heat sealing. The bag containing the element shall be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional, and the box closed and sealed as specified for method V in the appendix to the box specification. The box shall be sealed with tape conforming to PPP-T-60, type IV.

5.1.2 Extra O-rings, sight-gage gasket, and float-control gaskets. Each extra O-ring shall be coiled to the minimum practical diameter and each sight-gage gasket and each float-control gasket shall be laid flat and preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.3 Consolidation. The extra filter elements, O-rings, float-control gaskets, sight-gage gasket, and the float control and tubing shall be consolidated together in one or more boxes as applicable conforming to PPP-B-636, V3c, style optional. The contents shall be blocked, braced or cushioned as applicable to prevent movement or damage. Box closure and sealing shall be as specified for method V in accordance with the appendix to the box specification.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Each filter-separator and the consolidated components shall be packed together in a close-fitting box conforming to PPP-B-621, class 2, style 7, with hood. The filter-separator and the consolidated components shall be secured within the box in a manner to prevent movement and damage in accordance with MIL-STD-1186. The box shall be closed and strapped in accordance with the appendix to the box specification; strapping shall be zinc coated.

5.2.2 Commercial. Each filter-separator and the consolidated components (see 5.1.3) shall be packed together in a container in accordance with ASTM D 3951.

5.3 Marking (see 6.6).

5.3.1 Military. Marking for level A shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be shown on the shipping container.

6. NOTES

6.1 Intended use. The filter-separator is intended for use in airfield refueling systems, motor fuel servicing equipment, and military pipeline systems, for the removal of undissolved water and solid contaminants from aviation, diesel or motor fuels.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When the Government will conduct any or all of the first article examination and tests. Also, which additional first article tests will be conducted (see 3.2).
c. Color of finish paint, when other than specified (see 3.19).
d. When filter-separator shall be tested in accordance with 4.5.2.2, 4.5.2.4, 4.5.2.5, 4.5.2.7, 4.5.2.8, and 4.5.2.9 (see 4.3.2).
e. Degree of packing (see 5.2).

6.3 Classification change. The type I filter-separator with skids has been discontinued since it is no longer required. Since there is no technical difference between the type I and type II filter-separators, except for the skids, the requirement for classifying them is no longer needed.

6.4 First article. When a first article inspection is required, the items should be a production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.11.

6.6 Preservation/packing level marking. Degrees of preservation have not been included, since the protection specified is the minimum acceptable. For purposes of preservation/packing level markings, preservation shall be designated level A.

6.7 Subject term (key word) listing.

Fuel
Filter/separator
Filter elements

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Preparing activity:
Army - ME
Army - ME
Navy - AS
Air Force - 99

Project 4330-0038

Review activities:
Army - AT, MR
Navy - YD
Air Force - 84

User activity:
Army - AV
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-52694D	Filter-Separator, Liquid Fuel: Frame-Mounted, 600 GPM Capacity (2271 Liter Per Minute)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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